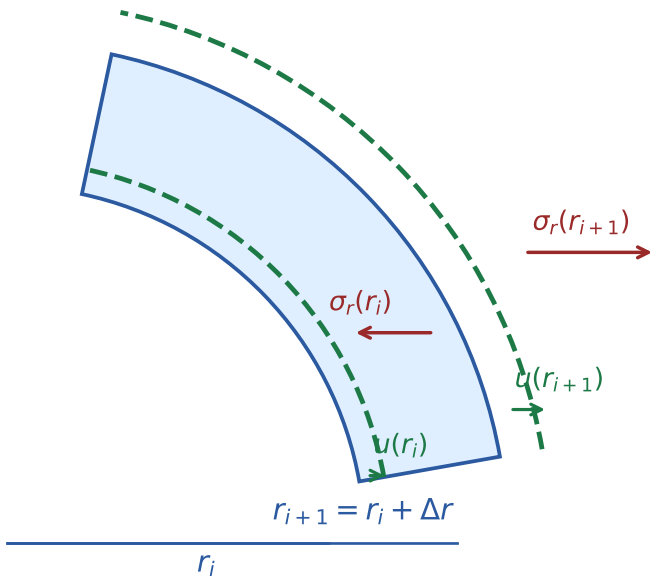


Dashed arcs: thermally expanded ring boundaries  
Different radial displacement across the ring drives strain compatibility and radial stress balance.



Compatibility:

$$\frac{d(r \varepsilon_\theta)}{dr} = \varepsilon_r$$

Radial equilibrium:

$$\frac{d(r \sigma_r)}{dr} = \sigma_\theta$$